Stephen Iota

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EDUCATION

M.S., Computer Science, University of Southern California 2023

B.S., Physics, University of California, Riverside 2019

EXPERIENCE

Graduate Student Researcher, *USC Viterbi School of Engineering* Jan 2021 — Dec 2022

- Developed safe and robust transfer/meta reinforcement learning methods through unsupervised causal representation learning, significantly outperforming extant baselines and publishing results as an AISTATS conference paper.
- Managed machine learning experiments on GPU servers, achieving faster results and higher accuracy through parallel processing and optimized hardware utilization.

Machine Learning Intern, *Information Science Institute* Summer 2021

- Implemented a cutting-edge machine learning pipeline to automate classification of geological data and to improve science workflow of Sonoma State Geology research group.
- Combined custom fine-tuned deep learning vision models with experiment tracking, visualization heatmaps and comparative metrics for easy exploration of different models and hyperparameters.
- Presented poster at "Proven AI/ML Applications in the Earth Sciences" session in AGU Fall Meeting.

Data Science Intern, *Information Science Institute* Jan 2021 — May 2021

- Conducted probabilistic modeling, numerical analysis and empirical validation of directed networks, useful for analyzing various phenomena in network science and computational social science, publishing work in Journal of Complex Networks.
- Demonstrated strong analytical, problem-solving, and critical thinking skills, as well as attention to detail and ability to work collaboratively in a research team.

Research Intern, *NASA Ames Research Center* Summer 2018

- Proposed new technique for more accurate and efficient detection and segmentation of 3D X-ray tomography data of NASA parachute materials, including combining nontrivial feature extraction with efficient detection and tracking algorithms.
- Integrated techniques into an existing data processing pipeline built to study and improve performance of parachute material under extreme conditions.

SKILLS

- Python, C/C++, Golang, Bash, LaTeX.
- Packages: PyTorch, JAX, NumPy, Pandas, Scikit-learn, Huggingface.
- Excellent written and verbal communication skills.
- Independently driven and easily adaptable in a dynamic environment.
- Software engineering best practices (e.g. version control, testing, code review, etc.).

OUTREACH & MEMBERSHIPS

Research Fellow, *USC AI Safety* Fall 2022
Team Lead, *USC Graduates Rising in Data Science (GRIDS)* 2021 — 2022
SI Leader (Physics and Math tutor), *UCR Academic Resource Center* 2018 — 2019

PUBLICATIONS

I. S. A. Sontakke*, **S. Iota***, Z. Hu*, A. Mehrjou, L. Itti and B. Schölkopf. "GalilAI: Out-of-Task Distribution Detection using Causal Active Experimentation for Safe Transfer RL." *In Proceedings of The 25th International Conference on Artificial Intelligence and Statistics,* Mar. 2022. arXiv, website, code.

II. B. Nettasinghe, N. Alipourfard, **S. Iota**, V. Krishnamurthy and K. Lerman. "Scale-free degree distributions, homophily and the glass ceiling effect in directed networks." *Journal of Complex Networks*, April 2022. DOI.